

Zingiber officinalis an herbal appetizer in the tiger shrimp *Penaeus monodon* (Fabricius) larviculture

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Abstract

Penaeus monodon postlarvae were fed with different percentages (0%, 25%, 50%, 75% and 100%) of the herbal appetizer *Zingiber officinalis* enriched *Artemia*. After 30 days of culture (i.e. PL-1–30), a very positive result was found in *Z. officinalis*-enriched *Artemia*-fed postlarvae. The unenriched *Artemia*-fed postlarvae consumed 91.0 mg/animal/30 days of feed, whereas the *Z. officinalis*-enriched *Artemia* increased their consumption to 127.9 mg/animal/30 days. A similar pattern was noticed in feed absorbed (110.2 mg), dry weight growth (26.7 mg) and feed catabolized (83.2 mg) in *Z. officinalis*-enriched *Artemia* because of enzymatic activities. The conversion efficiency of unenriched postlarva was 17.19%, whereas in 100% *Z. officinalis*-enriched *Artemia*, the maximum conversion efficiency was 20.85%. The net production efficiency increased significantly ($P < 0.05$) to 22% from that of the unenriched *Artemia*-fed postlarvae. The administration of *Z. officinalis* in all levels produced significantly ($P < 0.05$) higher weight gain and specific growth rate. The utilization efficiency of feed increased proportionately to the percentages of *Z. officinalis*. Digestive enzyme activity (amylase, protease and lipase) increased significantly ($P < 0.05$) in the 50%, 75% and 100% enrichment. Among the different percentages of enrichment, the 100% *Z. officinalis*-enriched *Artemia*-fed postlarvae performed better in the overall status.

KEY WORDS: *Artemia franciscana*, bioencapsulation, bioenergetics, growth promoter, herbal appetizer, larviculture, live feed, *Penaeus monodon*, *Zingiber officinalis*

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Introduction

Shrimp culture is considered as one of the lucrative industries because of the high market price and great demand in the international market (Varghese 1995). Sustained growth will depend on regular supply of adequate quantities of seeds that ensure high growth rate, low mortality rate and ability to withstand stress in high-density culture (Santhanakrishnan & Viswakumar 1995). To improve the quality of shrimp, feed consumption and sufficient nutrients to the larvae are important factors. Hormones, vitamins, antibiotics and other synthetic drugs have been tested for various activities, such as appetizing, growth promotion, immunostimulant and so on in aquaculture (Jayaprakas & Sambhu 1996). Even though some synthetic products promote larval production of fishes and shrimps (Sambhu 1996), they are not much preferred in commercial aquacultural operations because of their cost, tendency to form residues and undesirable side effects. Consumer awareness and concern over food safety has led to the search for alternative appetizers of natural origin. Plants are the storehouses and rich sources of safer and cheaper chemicals. Natural plant products serve as growth promoters, tonics, immunostimulants and appetizers (Nadkarni 1995). They are highly promising in the aquaculture industry to increase feed consumption and improve digestion, thereby leading to better growth and production. Ginger (*Zingiber officinalis*) belongs to Zingiberaceae family. The part of the plant used is the rhizome, an important spice.

The use of spices as food and feed additives has been practised widely since ancient times. Apart from enhancing the taste and flavour of food, spices have been widely believed to exert digestive stimulant action. Spices such as mint, garlic and ginger are the usual digestive stimulants. In animal models, especially rat, *Zingiber* plays a very important role in fat digestion and absorption (Bhat *et al.* 1985; Sambaiah &

Srinivasan 1991; Platel & Srinivasan 2000). Till date, no study has been carried out on the shrimp with *Z. officinalis* as an herbal appetizer. Therefore, in the present study, *Z. officinalis* was chosen and the stimulatory effect verified. Various percentages of *Z. officinalis* was prepared and fed to postlarvae (PL-1–30) of *Penaeus monodon* through the live feed *Artemia franciscana*, because of its versatile characteristics, such as taste, high nutritive value, nonselecting filter-feeding capability and non-contamination of the culture water.

Materials and methods

Rhizomes of *Z. officinalis* was shade-dried and ground into a fine powder (particle size $<50\ \mu\text{m}$). Different (enrichment) diets were prepared using boiled rice bran ($<50\ \mu\text{m}$ size) and *Z. officinalis* powder at different concentrations of 25%, 50%, 75% and 100% (no rice bran).

The mysis stage of *P. monodon* were purchased from a commercial hatchery in Kanyakumari, Tamil Nadu, India. They were carefully transported to the laboratory in oxygenated polythene bags. These mysis were maintained in laboratory conditions (room temperature $29 \pm 2\ ^\circ\text{C}$, dissolved O_2 $6.1 \pm 1.0\ \text{mg/L}^{-1}$) in polyvinyl chloride tanks and reared till the postlarval stage (PL-1). Two types of experiments were carried out. Experiment I was performed to assess the survival rate and digestive enzyme activities (amylase, protease and lipase). Postlarvae (PL-1) with an average mean weight of $6.73 \pm 0.1\ \text{mg}$ were stocked in every 15-L tank at the rate of 10 PL-1. Three replicates were maintained in each enrichment (25%, 50%, 75% and 100% of *Z. officinalis*). All tanks were connected through a continuous water flow system. Mild aeration was applied continuously to maintain the O_2 level. The values of mean salinity ($30.8 \pm 1.1\ \text{g L}^{-1}$), temperature ($23.9 \pm 1.5\ ^\circ\text{C}$), pH (8.1 ± 0.3), dissolved oxygen ($5.2 \pm 0.7\ \text{mg L}^{-1}$) and ammonia ($0.008 \pm 0.01\ \text{mg L}^{-1}\ \text{NH}_4^-\text{N}$) were recorded during the culture period using the standard method (APHA 1995). The debris, moulted shell and dead larvae were removed daily.

Experiment II was performed for bioenergetic studies. Postlarvae (PL-1) were stocked in a series of small glass aquaria (1 L capacity) at the rate of two per aquarium. Before introducing postlarvae, average mean length and weight were measured, five replicates were maintained and 80% of water was exchanged daily. Faeces and uneaten feed were collected carefully and dried in a hot air oven ($50\ ^\circ\text{C}$). Bioenergetics was estimated using the International Biological Program (IBP) formula following the standard method (Petrusewicz & MacFadyen 1990; Marian & Murugadas

1991). Specific growth rate (SGR) and feed conversion ratio (FCR) were calculated according to the formulas: $\ln W_2 - \ln W_1 / t_2 - t_1$ and $W_{\text{dr}} / (W_2 - W_1)$ respectively, where W_1 and W_2 are the initial and final weights of larvae, W_{dr} is the dry weight of feed and t is the time in days.

Feed consumption (C) was estimated gravimetrically by subtracting the dry weight of uneaten feed from the dry weight of the feed supplied. Feed absorbed (A) was estimated by subtracting the dry weight of the faeces (F) from that of feed consumed ($A = C - F$). Production (dry weight growth) was determined by subtracting dry weight of the larvae at the commencement of experiment from the dry weight of the larvae at termination of experiment. Feed catabolized (R) was estimated by subtracting the sum of tissue production from the feed absorbed ($R = A - P$). Net weight gain was calculated by subtracting initial wet weight from the final wet weight. Absorption efficiency was calculated by relating feed absorbed to feed consumed,

$$\text{Absorption efficiency} = \frac{\text{Feed absorbed}}{\text{Feed consumed}} \times 100$$

Gross and net conversion efficiencies were calculated using the following formulas

$$\text{Gross conversion efficiency} = \frac{\text{Feed converted}}{\text{Feed consumed}} \times 100$$

$$\text{Net conversion efficiency} = \frac{\text{Feed converted}}{\text{Feed absorbed}} \times 100$$

Artemia franciscana (San Francisco Bay brand) nauplii (to feed earlier postlarvae) and juveniles (to feed late larvae) were enriched with the different percentages of *Z. officinalis* (25%, 50%, 75% & 100%) for 48 h at a concentration of $400\ \text{mg L}^{-1}$. This is regarded optimum for herbal fine powders (Citarasu 2000). The postlarvae of experiment I and II received enriched and unenriched *Artemia* in an *ad libitum* feeding regime at different intervals.

The influence of *Z. officinalis* on digestive enzyme (amylase, protease and lipase) activities (total and specific) in the digestive tract was estimated by the method of King (1965). For preparation of enzyme extracts, postlarvae were starved for 24 h immediately after termination of the experiment and killed by pithing. The digestive tracts were carefully removed, thoroughly washed with sterile distilled water, weighed and homogenized with chilled saline (0.65%). The supernatant, extracted by centrifugation, was used for enzyme assays. Unit amylase activity was calculated as the weight (mg) of maltose liberated for a duration of 10 min at $30\ ^\circ\text{C}$. Unit protease activity was expressed as the amount of tyrosine liberated in 15 min under the assay conditions. Unit lipase

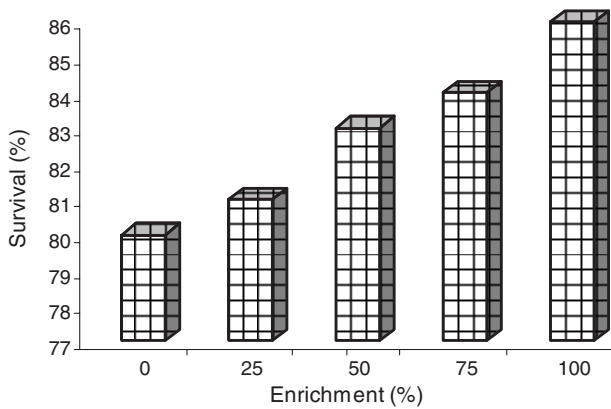


Figure 1 Graph shows the survival (%) versus *Z. officinalis* enrichment (%).

activity was expressed as the amount of 0.025 N NaOH required to neutralize the fatty acids liberated during 18 h of incubation at pH 6.9 and temperature 30 °C. Total and specific activities of digestive enzymes were calculated as enzyme unit per gram tissue and enzyme unit per milligram protein, respectively. The data obtained were analysed statistically through Tukey's test (Zar 1974).

Results

Postlarvae of *P. monodon* showed 80% survival when the live feed was not enriched with *Z. officinalis*. When the concentration of *Z. officinalis* was raised in the live feed, the survival rate also increased proportionately. Maximum survival (86%) was found in the 100% enrichment (Fig. 1). Consumption of 91.03 mg/animal/30 days was observed when the postlarvae were fed with unenriched *Artemia*, whereas *Z. officinalis* enrichment significantly ($P < 0.05$) increased consumption at all concentrations with a maximum of

127.9 mg/animal/30 days in the 100% treatment. The same trend was detected in absorption. The lowest production (growth) was observed (15.6 mg/animal/30 days) in the unenriched *Artemia*-fed postlarvae. But there was significant ($P < 0.05$) increases of 18.0, 21.1, 24.2 and 26.7 mg/animal/30 days in the 25%, 50%, 75% and 100% *Z. officinalis*-enriched *Artemia*-fed postlarvae, respectively. The minimum amount of feed catabolized (63 mg) was recorded in control (unenriched with *Artemia*) and the maximum amount of feed catabolized (83.2 mg) was observed in 100% *Z. officinalis*-enriched *Artemia*-fed postlarvae. The lowest (17.2%) gross conversion efficiency was observed in the unenriched *Artemia*-fed postlarvae. This significantly increased to 18.2%, 20.0% and 20.9% in 50%, 75% and 100% *Z. officinalis*-enriched *Artemia*-fed postlarvae. The net production efficiency reflects the same pattern as that of gross conversion efficiency. Highest FCR value (1.21) was observed in the unenriched *Artemia*-fed postlarvae. FCR values significantly decreased to 1.11%, 1.07%, 1.01% and 0.97% in the 25%, 50%, 75% and 100% *Z. officinalis*-enriched *Artemia*-fed postlarvae, respectively (Table 1).

The lowest weight gain (74.8 mg/animal/30 days) was observed in unenriched *Artemia*-fed postlarvae. This was increased significantly ($P < 0.05$) to 90.3, 105.3, 118.0 and 130.8 mg/animal/30 days in 25%, 50%, 75% and 100% *Z. officinalis*-enriched *Artemia*, respectively. The SGR was lowest (12.1%) in unenriched and highest (13.9) in 100% *Z. officinalis*-enriched *Artemia*-fed postlarva (Table 2).

Amylase and protease (total) activities significantly ($P < 0.05$) increased in the enriched diets (all concentrations). But for specific amylase and protease activities, only 75% and 100% enrichment significantly ($P < 0.05$) varied. Total lipase activity at 75% and 100% enrichment alone significantly ($P < 0.05$) increased. No difference was measured in specific activity of lipase (Table 3).

Table 1 Bioenergetic parameters of *Panaeus monodon* postlarvae (PL-1–30) fed different percentages of *Zingiber officinalis*-enriched *Artemia*

Parameters	Treatments (%)				
	Control	25	50	75	100
Consumption (mg)	91.0 ^a ± 0.6	101.4 ^b ± 1.9	113.0 ^c ± 1.6	121.0 ^d ± 2.1	127.9 ^e ± 2.9
Absorption (mg)	78.6 ^a ± 0.7	87.6 ^{ba} ± 1.4	97.3 ^{cb} ± 1.0	103.9 ^{dc} ± 1.5	110.2 ^{ed} ± 1.8
Dry weight growth (mg)	15.6 ^a ± 0.3	18.0 ^d ± 0.4	21.1 ^e ± 0.8	24.2 ^d ± 0.7	26.7 ^e ± 0.9
Feed catabolized (mg)	63.0 ^a ± 0.4	69.8 ^b ± 1.0	76.3 ^c ± 0.2	79.8 ^d ± 1.1	83.2 ^e ± 0.9
Absorption efficiency (%)	86.3 ^a ± 0.3	86.5 ^{ba} ± 0.4	86.1 ^{cb} ± 0.3	85.9 ^{dca} ± 0.3	86.2 ^{edcb} ± 0.1
Gross conversion efficiency (%)	17.2 ^a ± 0.3	17.8 ^a ± 0.1	18.6 ^{bc} ± 0.5	20.0 ^d ± 0.3	20.9 ^{ed} ± 0.4
Net production efficiency (%)	19.8 ^a ± 0.3	20.5 ^{ba} ± 0.1	21.6 ^{cba} ± 0.6	23.3 ^{dc} ± 0.5	24.2 ^{ed} ± 0.5
Feed conversion ratio	1.2 ^a ± 0.0	1.1 ^b ± 0.0	1.1 ^c ± 0.0	1.0 ^d ± 0.0	0.97 ^e ± 0.05

Values with the same superscript do not differ from each other ($P < 0.05$) – Tukey's test.

Table 2 Growth parameters of the postlarvae of *Panaeus monodon* (PL-1–30) fed with different percentages of *Zingiber officinalis*-enriched *Artemia*

Parameters	Treatments (%)				
	Control	25	50	75	100
Initial length (mm)	6.7 ± 0.3	6.6 ± 0.4	6.8 ± 0.3	6.8 ± 0.4	6.8 ± 0.3
Final length (mm)	25.5 ± 0.5	27.3 ± 0.8	28.7 ± 0.8	29.5 ± 1.0	31.0 ± 0.5
Initial wet weight (mg)	2.0 ± 0.0	2.0 ± 0.3	2.0 ± 0.2	2.1 ± 0.2	2.1 ± 0.1
Final wet weight (mg)	77.4 ± 2.0	92.3 ± 3.1	107.3 ± 3.9	120.1 ± 4.5	132.9 ± 1.8
Net weight gain (mg)	74.8 ^a ± 2.0	90.3 ^b ± 3.2	105.3 ^c ± 3.7	118.0 ^d ± 4.5	130.8 ^e ± 1.7
Weight gain (%)	97.3 ± 0.1	97.8 ^{ba} ± 0.4	98.1 ^{cb} ± 0.1	98.2 ^{dc} ± 0.1	98.4 ^{cdb} ± 0.1
Specific growth rate	0.12 ± 0.01	0.138 ^{ba} ± 0.01	0.13 ^{cb} ± 0.01	0.14 ^{dc} ± 0.01	0.14 ^{cdc} ± 0.01

Values with the same superscript do not differ from each other ($P < 0.05$) – Tukey's test.

Table 3 Digestive enzyme activities in *Panaeus monodon* postlarvae (PL-1–30) fed different percentages of *Zingiber officinalis*-enriched *Artemia*

Enzyme	Activity	Enrichment				
		Control	25	50	75	100
Amylase	Total	32.3 ^a ± 4.5	334.3 ^b ± 5.1	353.7 ^e ± 4.2	363.66 ± 5.68	373.0 ^e ± 7.0
	Specific	7.4 ^a ± 0.3	8.06 ^{ba} ± 0.1	8.2 ^{cba} ± 0.1	9.1 ^{dcba} ± 0.1	10.1 ^{edcb} ± 0.2
Protease	Total	274.0 ^a ± 6.2	289.0 ^b ± 3.6	290.7 ^{cb} ± 10.0	320.0 ^d ± 5.0	345.313 ^e ± 5.7
	Specific	3.7 ^a ± 0.1	3.9 ^{ba} ± 0.0	4.0 ^{cb} ± 0.1	4.1 ^{dc} ± 0.1	4.2 ^{edcb} ± 0.1
Lipase	Total	3.8 ^a ± 0.1	3.9 ^{ba} ± 0.1	4.3 ^{bca} ± 0.3	4.7 ^{dca} ± 0.2	5.1 ^{dea} ± 0.1
	Specific	1.7 ^a ± 0.1	1.9 ^{ba} ± 0.1	2.2 ^{cba} ± 0.1	2.3 ^{dcba} ± 0.4	2.5 ^{edcba} ± 0.2

Values with the same superscript do not differ from each other ($P < 0.05$) – Tukey's test.

Discussion

The results of the present study clearly indicate that *Z. officinalis* plays a positive role on the feed consumption, growth, SGR, FCR and activities of amylase and protease. Our results are comparable with those of Livol (IHF-1000) – which is a herbal growth promoter containing different plant ingredients, such as *Boerhavia diffusa*, *Solanum nigrum*, *Terminalia arjuna*, *Colosynth* and black salt – and has been found to significantly improve digestion, thereby leading to better growth, production and health in cultivable fishes (Shadakshari 1993 and Jayaprakas & Euphrasia 1997). Similar positive observations were reported by Citarasu *et al.* (1998), with increased survival, growth and consumption in the postlarvae (PL-1–20) of *P. indicus* fed with the herbal products, stresstol-II- and stresstol-I-enriched *Artemia*, and also found that, stresstol which is a larval herbal diet, increased the feed consumption rate from 165.7 to 171.9 mg g day⁻¹, with a production rate from 31.5 to 42.4 mg g day⁻¹ and SGR from 7.3% to 9.1% when compared with the unenriched *Artemia*-fed postlarvae. In another study, Citarasu *et al.* (2002) reported that postlarvae feeding on *Artemia* enriched with a herbal medicinal diet, tolerated stress efficiently and showed increased SGR and maximum survival (95%).

Digestive enzyme activities in 75% and 100% *Z. officinalis*-enriched *Artemia*-fed postlarvae were highly increased, probably, because of the high concentration of compounds, such as zingiberene, gingerol, camphene, etc., also leading to increased feed consumption. Similar results were found in Livol-incorporated diets fed to fishes (Shadakshari 1993). Papaya leaf meal containing papain, when fed to *P. monodon* postlarvae, increased protein digestion, FCR, SGR and weight gain (Penaflorida 1995). Nutripro-aqua, a soya-based herbal growth promoter enhanced the digestive enzyme activity and improved growth in *Macrobrachium rosenbergii* (Keshavanth & Jayaram 2000). Different spices such as curcumin, fenugreek, ginger and ajowan significantly increased pancreatic secretions – protease, amylase and lipase in rat (Platel *et al.* 2002; Platel & Srinivasan 2004). *Zingiber officinalis* has shown significant stimulation of pancreatic amylase activity and also increases intestinal lipase activity in rats (Platel & Srinivasan 1996). Generally, all spices shorten the feed transit time; this reduction was more prominent in the case of *Z. officinalis*. The reduction in transit time might have a beneficial influence on digestive enzymes and could accelerate the overall digestive process (Platel & Srinivasan 2004). The enhanced proteolytic and lipolytic activities in the digestive tract of the *Z. officinalis*-

fed postlarvae could also be linked to better protein and lipid digestibilities.

In conclusion, *Z. officinalis* stimulated appetite in the post-larvae, leading to enhanced feed consumption, FCR, SGR and improved amylase and protease activities. These factors are all important to produce high-quality larvae. Adding specific ingredients to feed might improve feed intake and thereby reduce feed costs. Our results show that *Z. officinalis* is a new feed additive which can be successfully used in shrimp feed.

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